

A Technical Introduction to Policy gradient method

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a | s) Q^{\pi}(s, a)]$$

Section 1

$\gamma t (\sum_{k=0}^{n-1} \gamma^k R_{t+k} + \gamma^n V^{\pi_{\theta}}(S_{t+n}) - V^{\pi_{\theta}}(S_t))$ $\{\text{textstyle } \gamma \sum_{k=0}^{n-1} \gamma^k R_{t+k} + \gamma^n V^{\pi_{\theta}}(S_{t+n}) - V^{\pi_{\theta}}(S_t)\}$: n-step TD learning.

Section 2

■ $\{\text{displaystyle } \epsilon\}$ is the trust region radius. Note that in general, other surrogate advantages are possible: $L(\theta, \theta_i) = \mathbb{E}_{s, a \sim \pi_{\theta_i}} [\pi_{\theta}(a | s) \pi_{\theta_i}(a | s) \Psi \pi_{\theta_i}(s, a)]$ $\{\text{displaystyle } L(\theta, \theta_i) = \mathbb{E}_{s, a \sim \pi_{\theta_i}} [\pi_{\theta}(a | s) \pi_{\theta_i}(a | s) \Psi \pi_{\theta_i}(s, a)]\}$ where Ψ $\{\text{displaystyle } \Psi\}$ is any linear sum of the previously mentioned type. Indeed, OpenAI recommended using the Generalized Advantage Estimate, instead of the plain advantage $A^{\pi_{\theta}}(s, a)$ $\{\text{displaystyle } A^{\pi_{\theta}}(s, a)\}$. The surrogate advantage $L(\theta, \theta_i)$ $\{\text{displaystyle } L(\theta, \theta_i)\}$ is designed to align with the policy gradient $\nabla_{\theta} J(\theta)$ $\{\text{displaystyle } \nabla_{\theta} J(\theta)\}$. Specifically, when $\theta = \theta_i$ $\{\text{displaystyle } \theta = \theta_i\}$, $\nabla_{\theta} L(\theta, \theta_i)$ $\{\text{displaystyle } \nabla_{\theta} L(\theta, \theta_i)\}$ equals the policy gradient derived from the advantage function:

Section 3

$\gamma t \sum_{n=1}^{\infty} \lambda^{n-1} (1 - \lambda) \cdot (\sum_{k=0}^{n-1} \gamma^k R_{t+k} + \gamma^n V^{\pi_{\theta}}(S_{t+n}) - V^{\pi_{\theta}}(S_t))$ $\{\text{textstyle } \gamma \sum_{n=1}^{\infty} \lambda^{n-1} (1 - \lambda) \cdot (\sum_{k=0}^{n-1} \gamma^k R_{t+k} + \gamma^n V^{\pi_{\theta}}(S_{t+n}) - V^{\pi_{\theta}}(S_t))\}$: TD(λ) learning, also known as GAE (generalized advantage estimate). This is obtained by an exponentially decaying sum of the n-step TD learning ones.

Section 4

Policy gradient The REINFORCE algorithm, introduced by Ronald J. Williams in 1992, was the first policy gradient method. It is based on the identity for the policy gradient $\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} [\sum_{t=0}^T \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \sum_{\tau=t}^T (\gamma^{\tau-t} R_{\tau}) | S_0 = s_0]$ $\{\text{displaystyle } \nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} [\sum_{t=0}^T \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \sum_{\tau=t}^T (\gamma^{\tau-t} R_{\tau}) | S_0 = s_0]\}$ which can be improved via the "causality trick" $\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} [\sum_{t=0}^T \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \sum_{\tau=t}^T (\gamma^{\tau-t} R_{\tau}) | S_0 = s_0]$ $\{\text{displaystyle } \nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} [\sum_{t=0}^T \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \sum_{\tau=t}^T (\gamma^{\tau-t} R_{\tau}) | S_0 = s_0]\}$

$$\mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \sum_{\tau=t}^T (\gamma^{\tau-t} R_{\tau}) | S_0 = s_0] = \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \sum_{\tau=t}^T (\gamma^{\tau-t} R_{\tau}) | S_0 = s_0]$$

Section 5

The Group Relative Policy Optimization (GRPO) is a minor variant of PPO that omits the value function estimator V $\{\text{displaystyle } V\}$. Instead, for each state s $\{\text{displaystyle } s\}$, it samples multiple actions $a_1, \dots, a_G$ $\{\text{displaystyle } a_1, \dots, a_G\}$ from the policy π_{θ} $\{\text{displaystyle } \pi_{\theta}\}$, then calculate the group-relative advantage $A^{\pi_{\theta}}(s, a_j) = r(s, a_j) - \mu - \sigma$ $\{\text{displaystyle } A^{\pi_{\theta}}(s, a_j) = r(s, a_j) - \mu - \sigma\}$ where μ, σ $\{\text{displaystyle } \mu, \sigma\}$ are the mean and standard deviation of $r(s, a_1), \dots, r(s, a_G)$ $\{\text{displaystyle } r(s, a_1), \dots, r(s, a_G)\}$. That is, it is the standard score of the rewards. Then, it maximizes the PPO objective, averaged over all actions: $\max_{\theta} \frac{1}{G} \sum_{i=1}^G \mathbb{E}_{s, a_1, \dots, a_G} [\min(\pi_{\theta}(a_i | s) \pi_{\theta_i}(a_i | s), 1 + \epsilon) A^{\pi_{\theta}}(s, a_i) \text{ if } A^{\pi_{\theta}}(s, a_i) > 0, \max(\pi_{\theta}(a_i | s) \pi_{\theta_i}(a_i | s), 1 - \epsilon) A^{\pi_{\theta}}(s, a_i) \text{ if } A^{\pi_{\theta}}(s, a_i) < 0]$ $\{\text{displaystyle } \max_{\theta} \frac{1}{G} \sum_{i=1}^G \mathbb{E}_{s, a_1, \dots, a_G} [\min(\pi_{\theta}(a_i | s) \pi_{\theta_i}(a_i | s), 1 + \epsilon) A^{\pi_{\theta}}(s, a_i) \text{ if } A^{\pi_{\theta}}(s, a_i) > 0, \max(\pi_{\theta}(a_i | s) \pi_{\theta_i}(a_i | s), 1 - \epsilon) A^{\pi_{\theta}}(s, a_i) \text{ if } A^{\pi_{\theta}}(s, a_i) < 0]\}$ Intuitively, each policy update step in GRPO makes the policy more likely to respond to each state with an action that performed relatively better than other actions tried at that state, and less likely to respond with one that performed relatively worse. As before, the KL penalty term can be applied to encourage the trained policy to stay close to a reference policy. GRPO was first proposed in the context of training reasoning language models by researchers at DeepSeek.

Section 6

The natural policy gradient method is a variant of the policy gradient method, proposed by Sham Kakade in 2001. Unlike standard policy gradient methods, which depend on the choice of parameters θ $\{\text{displaystyle } \theta\}$ (making updates coordinate-dependent), the natural policy gradient aims to provide a coordinate-free update, which is geometrically "natural".

Section 7

$F = \nabla_{\theta}^2 D^{-1} \text{KL}(\pi_{\theta} || \pi_{\theta_i}) |_{\theta = \theta_i}$ $\{\text{displaystyle } F = \nabla_{\theta}^2 D^{-1} \text{KL}(\pi_{\theta} || \pi_{\theta_i}) |_{\theta = \theta_i}\}$ is the Fisher information matrix. This reduces the problem to a quadratic optimization, yielding the natural policy gradient update:

Section 8

If $b_i(S_t)$ is chosen well, such that $b_i(S_t) \approx \sum_{\tau=t}^T (\gamma^\tau R_\tau) = \gamma \sum_{\tau=t}^T \gamma^{\tau-t} V^{\pi_{\theta_i}}(S_\tau)$, this could significantly decrease variance in the gradient estimation. That is, the baseline should be as close to the value function $V^{\pi_{\theta_i}}(S_t)$ as possible, approaching the ideal of: $\nabla_{\theta} J(\theta) = E_{\pi_{\theta}} [\sum_{t=0}^T \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) (\sum_{\tau=t}^T (\gamma^\tau R_\tau) - \gamma V^{\pi_{\theta}}(S_t)) | S_0 = s_0]$. Note that, as the policy π_{θ_t} updates, the value function $V^{\pi_{\theta_i}}(S_t)$ updates as well, so the baseline should also be updated. One common approach is to train a separate function that estimates the value function, and use that as the baseline. This is one of the actor-critic methods, where the policy function is the actor and the value function is the critic. The Q-function Q^{π} can also be used as the critic, since $\nabla_{\theta} J(\theta) = E_{\pi_{\theta}} [\sum_{0 \leq t \leq T} \gamma^t \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \cdot Q^{\pi_{\theta}}(S_t, A_t) | S_0 = s_0]$. Subtracting the value function as a baseline, we find that the advantage function $A^{\pi}(S, A) = Q^{\pi}(S, A) - V^{\pi}(S)$ can be used as the critic as well: $\nabla_{\theta} J(\theta) = E_{\pi_{\theta}} [\sum_{0 \leq t \leq T} \gamma^t \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \cdot A^{\pi_{\theta}}(S_t, A_t) | S_0 = s_0]$. In summary, there are many unbiased estimators for $\nabla_{\theta} J(\theta)$, all in the form of: $\nabla_{\theta} J(\theta) = E_{\pi_{\theta}} [\sum_{0 \leq t \leq T} \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \cdot \Psi_t | S_0 = s_0]$ where Ψ_t is any linear sum of the following terms: